

Testing of Mobile Applications

Meenakshi D'Souza

International Institute of Information Technology Bangalore.

Overview

- Introduction to mobile apps testing.
- Key challenges to address.
- Types of mobile apps testing.
- Some emulators and tools.

NPTEL

Mobile apps testing

- Mobile application testing is a process by which application software developed for handheld mobile devices is tested for its functionality, usability and consistency.
- Can be automated or manual.
- Mobile applications either come pre-installed or can be installed from mobile software distribution platforms.
- Related area: Testing of wearable applications.

NPTEL

Key challenges for mobile applications testing

- Device fragmentation: Wide range of mobile platforms/operating systems.
  - Each operating system has its own limitations.
  - Each application has to be testing across multiple devices running on the same platform. This has to be repeated for several platforms.
  - Wide range of mobile devices with different screen sizes and hardware configurations.
  - Gaining access to various devices for testing is expensive and time consuming.

NPTEL

Key challenges for mobile applications testing

- Large number of mobile network operators.
  - It is estimated that there are over 400 mobile network operators in the world.
  - Some of them are common standards like GSM and CDMA, a large number of them are on less common standards.
  - Each network operator uses a different kind of network infrastructure, limiting the flow of information.

NPTEL

Key challenges for mobile applications testing

- Executing a test script across devices is a challenge.
  - Devices differ in key strokes, input methods, texture, display etc. Single script will not function on every device.
- Mobile apps have to be tested for *downloadability* from a particular store, across networks and devices.
- Compatibility across devices, platforms and networks is another key feature for testing.

NPTEL

Mobile application vs. desktop application testing

- On desktop, the application is tested on a CPU. On a mobile device, the application is tested on mobile handsets.
- Mobile device screen size is smaller than desktop.
- Mobile devices have less memory than desktop.
- Mobiles use network connections like 2G, 3G, 4G or WIFI whereas desktops use broadband or dial up connections.
- The automation tool used for desktop application testing might not work on mobile applications.

NPTEL

Types of mobile applications testing

To address the new challenges in mobile apps testing, there are several different types of mobile apps testing, some of them just tuned for mobile apps.

- Functional testing: Functional testing ensures that the application is working as per the requirements. Most of the tests conducted for this are driven by the user interface and call flow.
- Usability testing: Testing done to make sure that the mobile app is easy to use and provides a satisfactory user experience to customers.
  - Key to an app's commercial success.
  - Tests for uniform user experience across devices; addresses the device fragmentation issues.
  - Tests for functionality, placement/sizing across different devices.

Types of mobile applications testing

- Performace testing: Testing process done to check the performance and behavior of the application under certain conditions including
  - low battery, bad network coverage, low available memory, simultaneous access to applications server by several users etc.
- Performance testing is done on the client side and on the server side.
- Compatibility testing: Testing of the application in different mobiles devices, browsers, screen sizes and OS versions etc., according to the requirements.
  - Includes testing for finer aspects like screen resolution, local language features etc.

Types of mobile applications testing

- Memory-related testing: This testing is done to ensure that the application manages memory and doesn't slow down the overall system.
  - Mobile devices have significant constraints on memory.
  - Applications should delete temporary files, use screens and other hardware resources optimally (*wakelocks*) etc.
- Installation testing:
  - Some applications come pre-installed on a devices, others have to be downloaded and installed.
  - This testing covers installation, updating and uninstalling of an application.

NPTEL

Types of mobile applications testing

- Interrupt testing: An application may face interrupts while installing or while in use.
- Interrupts include
  - Incoming and outgoing SMS and MMS, incoming and outgoing calls, incoming notifications, cable insertion and removal for data transfer, network outage and recovery, media Player on/off, device power cycle etc.
- Interrupt testing tests if an application correctly goes into a suspended state and resumes afterwards.

NPTEL

Types of mobile applications testing

- Security testing: Several mobile phones are used for secure transactions including bank transactions, credit card payments etc.
- Security testing checks for vulnerabilities involving hacking, access control and authentication, data security, session management etc.
- There are several security standards like Common Criteria (CC), NIST standard etc. that need to be adhered to.
- This is a very involved area, not covering any depth here.
  - Different kinds of attacks: Hardware, network, application software.
  - Program analysis, coding standards etc. play an important role.
  - Extensive cryptography.

Types of mobile applications testing

- Location testing: Typically, connectivity changes with network and location. Functionality of an application needs to be tested for these changes.
- Very difficult to mimic these changes in a laboratory/development center and test.

NPTEL

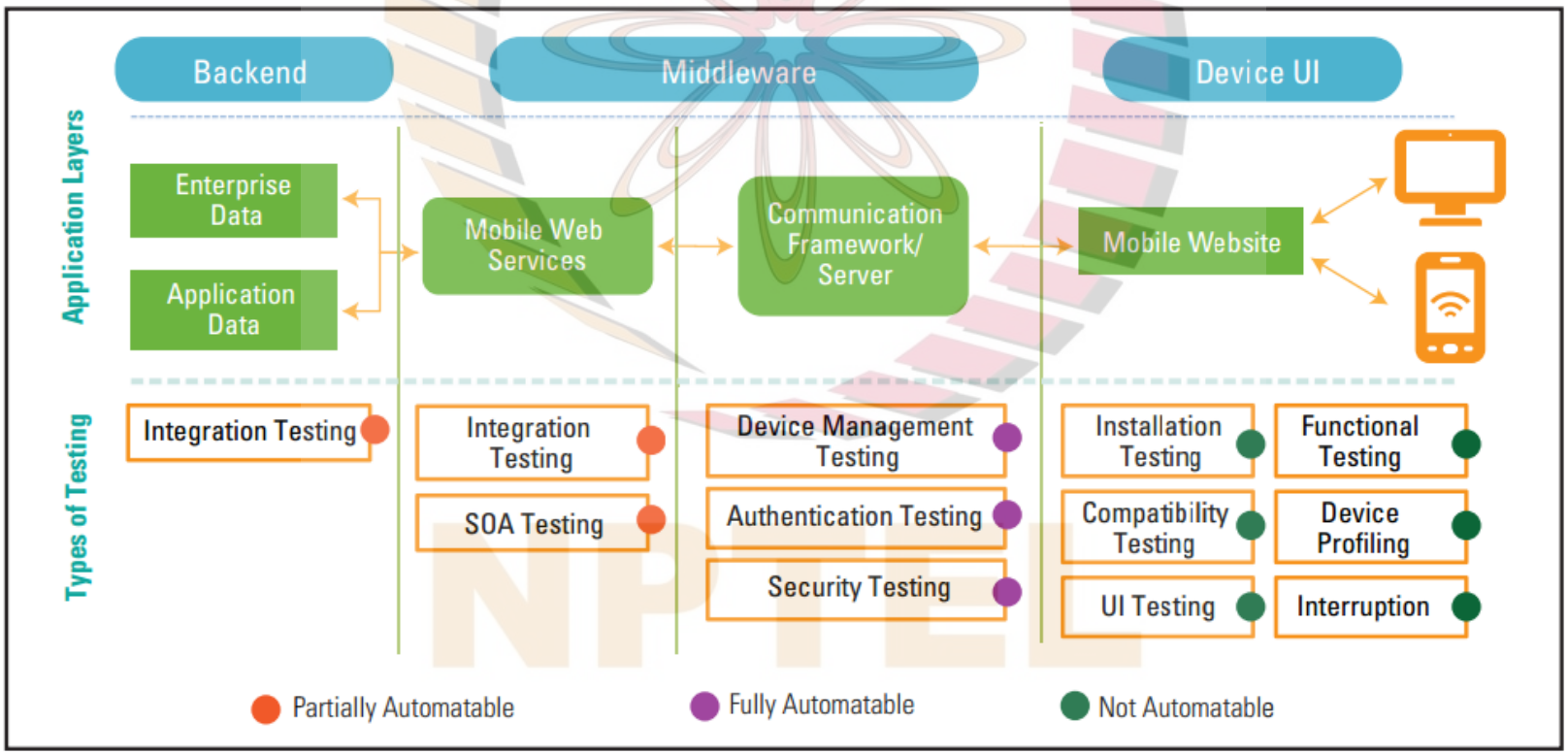
Types of mobile applications testing

- Apart from the listed system level features, normal unit testing, integration testing etc., are performed for mobile phone apps.
- These can be done using the criteria that we learnt in the course and using mutation testing.
- The listed system level testing are maunally done, cannot be fully automated.

NPTEL

Types of mobile applications testing

Summary slide on mobile application testing framework. Image credit: Software Testing class website.



Use of emulators in testing of mobile apps

- Typical system level features that are tested for mobile apps are tested using emulators.
- Emulator is a system/software that runs a software from one environment (platform) in another environment without changing the software itself.
- Emulators work by *duplicating* the features of a real system in a software+system set-up.

NPTEL

Types of mobile emulators

- Device Emulator: Provided by device manufacturers.
- Browser Emulator: Simulates mobile browser environments.
- Operating systems Emulator: Provided by the OS firm (Apple provides emulators for iOS, Microsoft for Windows phones and Google for Android phones).

NPTEL

Some (free) mobile phone emulators

- Mobile Phone Emulator:
  - Used to test the display of a particular website on a cell phone terminal.
  - Available for use from `http://www.mobilephoneemulator.com/`.
- mobiReady ^{TM}:
  - A free tool for testing website performance on mobile devices.
  - Available for use from `https://ready.mobi/`.

NPTEL

Some other mobile apps testing tools

- Ranorex ®:
  - A GUI testing tool for mobile apps.
  - Available for free trial from `https://www.ranorex.com/`.
- Jamo Automator.
  - A scriptless cloud-based test automation tool for mobile apps testing.
  - Free trail available from `https://www.jamosolutions.com/automator-1`.

NPTEL

Mobile apps testing: Summary

- This lecture only gives an overview of mobile apps testing.
- Main focus is on system-level testing of specific quality attributes.
- Needs extensive tool support, specific domain knowledge and training.

NPTEL